SAR-Acoustic Localization: A Low-Cost Embedded Acoustic Source Localization System for Disaster Search and Rescue

Gökçer Erol† Hafize Ulya Topbaş* İbrahim Emek* Şeref Anzerli*
*Computer Engineering Department, Hacettepe University, Ankara, Türkiye
†Graduate School of Science and Engineering, Hacettepe University, Ankara, Türkiye
E-mails: †gokcererol25@hacettepe.edu.tr, *hfzulya@gmail.com, *emekibr02@gmail.com, *serefanzerli@gmail.com

***Abstract*—This paper proposes a distributed acoustic localization system designed to assist search and rescue teams in rapidly detecting and locating survivors trapped under debris following structural collapses. The system estimates the position of sound sources, such as whistle, utilizing the Time Difference of Arrival (TDoA) principle. Multiple embedded sensor nodes, synchronized at the microsecond level via GNSS-based Pulse-Per-Second (PPS) signals, capture acoustic events and transmit timestamp data via long-range, low-power LoRa wireless modules. A central edge computing gateway collects this data to calculate the sound source coordinates through multilateration algorithms. To ensure accuracy in complex disaster environments, the system employs Event-triggered timestamp TDoA over LoRa with geometric multilateration refinement for precise time delay estimation and localization, alongside lightweight machine learning models for acoustic event classification and noise filtering. The resulting portable prototype provides real-time position estimation visualized on a web-based dashboard, demonstrating the feasibility and effectiveness of employing affordable, distributed embedded hardware to accelerate critical disaster-response operations.**

***Index Terms*—Smart IoT, Acoustic Localization, Wireless Sensor Network (WSN), Disaster Search & Rescue Systems (SAR), Time Difference of Arrival (TDoA)**

I. Introduction

Natural disasters, particularly earthquakes and structural collapses, create complex environments where survivors are often trapped under dense layers of rubble. The "Golden 72 Hours" period is a critical window for search and rescue (SAR) operations, as the probability of survival decreases exponentially after this duration. Traditional search methods—such as rescue dogs or thermal imaging—often face limitations in terms of high deployment costs and signal attenuation through reinforced concrete. In these scenarios, acoustic signals produced by survivors, such as rhythmic knocking or blowing the whistle, offer a resilient alternative as they can propagate through solid debris. Research has shown that three-dimensional (3D) localization methods based on wireless sensor networks (WSN) can be effectively applied to search for earthquake survivors in these extreme conditions [1].

Implementing a distributed acoustic localization system on resource-constrained hardware presents several significant embedded system challenges that must be addressed to ensure operational reliability. First, achieving precise time synchronization across distributed sensor nodes is paramount for Time Difference of Arrival (TDoA) estimation, a method extensively surveyed for its suitability in wireless acoustic sensor networks [2]. Since even microsecond-level offsets can lead to significant localization errors, the system must handle the inherent complexity of TDoA-based microphone arrays [1]. Second, accurate timestamping of acoustic events in non-stationary environments is required. While previous studies have utilized platforms like Raspberry Pi for real-time estimation [3], the challenge remains to maintain high precision on even lower-power microcontrollers. Finally, performing real-time localization with noisy acoustic data requires advanced processing; recent developments suggest that integrating convolutional neural networks (CNN) on edge devices can significantly improve robustness against environmental noise [4].

This paper proposes a low-cost, distributed acoustic source localization platform designed to address these challenges. The system utilizes multiple ESP32-S3 sensor nodes equipped with digital I2S microphones for high-quality audio acquisition. To ensure synchronization, a GNSS-based Pulse-Per-Second (PPS) mechanism is implemented to provide the microsecond-level precision required for TDoA. Communication is handled via LoRa modules to ensure long-range connectivity under low-power constraints. Moreover, ESP32 utilizes lightweight machine learning model for noise filtering and multilateration. At the central gateway, a Jetson Nano platform executes event-triggered timestamp TDoA over LoRa with geometric multilateration refinement.

The primary contribution of this work is the integration of affordable embedded hardware with advanced signal processing to create a portable SAR tool. By demonstrating a real-time localization prototype with a web-based visualization dashboard, this project aims to bridge the gap between laboratory-grade equipment and practical, field-deployable disaster-response technology.

II. Related Work

Acoustic source localization (ASL) has been a focal point of research within wireless acoustic sensor networks (WASN), particularly for its potential in environments where traditional sensing is limited [2]. The existing literature can be catego-

rized into localization methodologies, hardware implementation strategies, and noise-robustness enhancements.

A. Localization Methodologies

Traditional ASL methods typically rely on Time Difference of Arrival (TDoA), Direction of Arrival (DoA), or Steered Response Power (SRP) [2]. TDoA-based approaches are particularly suitable for distributed embedded systems, as they primarily require timestamp information rather than high-bandwidth raw audio transmission [2]. Dong and Yu further explored TDoA-based microphone arrays, highlighting the critical role of precise time-delay estimation in achieving localization accuracy [1]. Specifically for post-disaster scenarios, Wang et al. proposed a 3D localization method for earthquake survivors based on WSN nodes, proving the feasibility of sound-based search in collapsed structures [5].

B. Hardware and Edge Implementation

Implementing these algorithms on low-cost hardware is a significant challenge in the literature. Patwari et al. demonstrated real-time DoA estimation using Raspberry Pi and microphone arrays, showing that consumer-grade platforms can handle complex signal processing tasks [3]. However, many such systems remain limited by short-range connectivity or high power consumption. Our work builds on these implementations by transitioning to the ESP32-S3 platform and utilizing LoRa for long-range communication.

C. Robustness and Machine Learning

Signal accuracy in reverberant and noisy disaster environments often degrades due to environmental interference. While Generalized Cross-Correlation with Phase Transform (GCC-PHAT) is a standard for time-delay estimation [1], recent research integrates deep learning for improved robustness. Yin and Verhelst proposed a CNN-based SRP-PHAT model designed for the "extreme edge," demonstrating that complex acoustic event classification can be performed on resource-constrained devices to filter out meaningful signals from noise [4].

TABLE I
COMPARISON OF SYSTEMS BASED ON SPECIFIC HARDWARE AND COMMUNICATION PROTOCOLS

Work	Computing Platform	Comm. Module	Methodology	Sync. Method
Wang et al. [5]	Generic WSN (MSP430)	ZigBee (802.15.4)	3D TDoA	WSN Sync
Dong & Yu [1]	High-Speed PC/DAQ	Wired (Multi-Ch)	TDoA Est.	Wired Clock
Cobos et al. [2]	Distributed WASN	Generic Wireless	TDoA/DoA/SRP	Network Based
Patwari et al. [3]	Raspberry Pi 4	WiFi / Local	Real-time DoA	Local Bus
Yin & Verhelst [4]	GAP8 (RISC-V)	N/A (On-chip)	CNN / SRP-PHAT	N/A
This Work	**ESP32-S3 & Jetson Nano**	**LoRa**	**TDoA + ML**	**GNSS PPS**

D. Positioning of the Proposed System

While the reviewed works provide high accuracy in controlled or laboratory settings, they often lack the integration of long-range communication and microsecond-level synchronization necessary for field-deployable search and rescue (SAR) operations. The proposed system improves upon existing methodologies by combining GNSS-based PPS synchronization [1] with LoRa connectivity and edge-AI robustness [4], creating a portable and affordable solution specifically tailored for disaster response.

III. PROBLEM DEFINITION

The technical objective is the precise estimation of a sound source's coordinates $P = (x_p, y_p)$ within a two-dimensional (2D) Euclidean plane using a distributed network of resource-constrained embedded nodes. Formally, given a set of N sensor nodes at known fixed locations $S_i = (x_i, y_i)$, the system must solve for P by minimizing the error in the non-linear multilateration equations derived from the Time Difference of Arrival (TDoA) of a detected acoustic event.

A. System / Application Description

The application focuses on post-disaster Search and Rescue (SAR) in environments such as collapsed structures or defined debris fields where a 2D approximation is geographically sufficient.

- **Sensor Nodes:** Four distributed nodes utilizing ESP32-S3 microcontrollers and I2S digital microphones for high-fidelity audio capture.
- **Synchronization:** Microsecond-level temporal alignment is achieved via GNSS-based Pulse-Per-Second (PPS) signals, creating a unified time-base across the LoRa star topology.
- **Central Processing:** Data is aggregated at an NVIDIA Jetson Nano gateway which performs the TDoA solving and event visualization.
- **Assumptions:** The propagation medium is assumed to be isotropic with a constant speed of sound ($v \approx 343$ m/s), and the survivor's signal is treated as a point source.

B. Objectives

The project aims to meet the following technical benchmarks to ensure operational viability:

- **Synchronization Precision:** Maintain a clock drift of less than 10 μs across nodes to ensure sub-meter localization accuracy, as timing errors are the primary source of TDoA failure.
- **Low-Latency Response:** Achieve a "signal-to-dashboard" latency of less than 2 seconds to allow rescue teams to react to transient survivor signals.
- **Robust Detection:** Implement lightweight AI models on the edge to achieve at least 85% classification accuracy, distinguishing human knocking from mechanical environmental noise.

C. Constraints

The system design is governed by the following embedded and environmental limitations:

- **Hardware Processing:** The ESP32-S3 must perform real-time I2S DMA sampling and peak detection within its available SRAM and clock cycles.
- **Communication Bandwidth:** Due to LoRa's low bitrate, the system is constrained to transmitting only 48-byte metadata packets (timestamps and regarding information about event) rather than raw audio streams.
- **Power Consumption:** Each node must operate for a minimum of 24 hours on a 3.7V LiPo battery, requiring efficient use of deep-sleep modes between acoustic triggers.

- **Environmental Interference:** The system mitigates multi-path effects and reverberation in rubble through a timestamp-based TDoA pipeline with solver-side robustness, including physical-consistency checks, GDOP/residual gating, anti-ghost filtering, and weighted multilateration (SNR/ML-confidence weighting), rather than PHAT-weighted cross-correlation.

IV. System Architecture and Design

The system is designed as a distributed wireless acoustic sensor network (WASN) utilizing a star topology. Four independent sensor nodes are deployed across the target area to capture acoustic events, while a central gateway acts as the primary computation hub for time-difference aggregation and coordinate estimation.

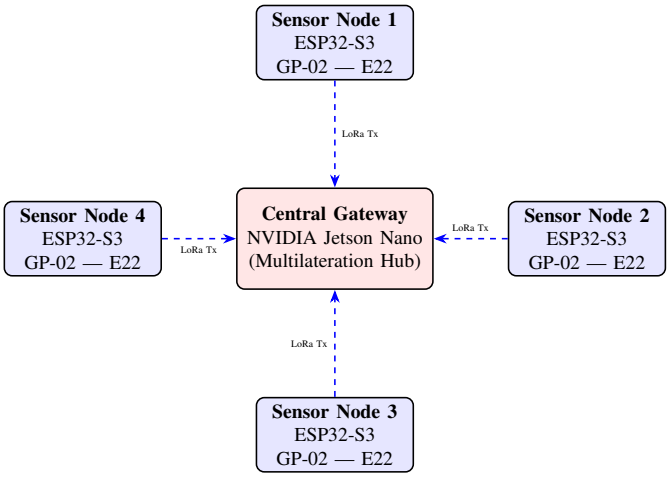

Fig. 1. System Architecture: Star topology featuring four distributed ESP32-S3 sensor nodes and a central NVIDIA Jetson Nano gateway.

A. Hardware/Software Architecture

Hardware Components:

- **Sensor Nodes (MCU):** The network consists of four nodes based on the **ESP32-S3** microcontroller. These nodes are distributed within the field to provide multiple perspectives of the acoustic source.
- **Synchronization:** Every node is equipped with a **Bordoza GP-02 GNSS** module. The Pulse-Per-Second (PPS) output of these modules provides a global time-base, allowing nodes to timestamp acoustic events with microsecond precision.
- **Communication: Ebyte E22-900T22D** LoRa modules facilitate long-range wireless data transfer from the nodes to the center, ensuring signal penetration through obstacles.
- **Gateway:** A central **NVIDIA Jetson Nano** hosts a matching E22 LoRa receiver. It processes the incoming asynchronous timestamps to compute the source location.

B. Dataflow and System Model

The system model follows a "detect-and-forward" dataflow where high-bandwidth audio processing is performed at the edge (nodes) and low-bandwidth metadata is processed at the core (gateway).

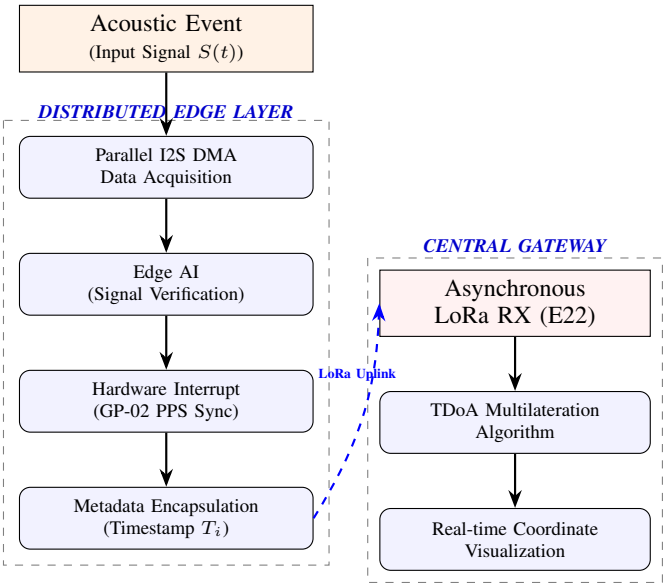

Fig. 2. Functional dataflow and system pipeline.

V. Requirements Analysis (Implementation Track)

The design of the distributed acoustic localization system is governed by a set of functional and non-functional requirements that ensure reliable performance in disaster-recovery scenarios.

A. Functional Requirements

1) **Distributed Signal Capture:** The system shall utilize four distributed sensor nodes to capture acoustic transients from multiple spatial perspectives, providing the redundancy required for robust TDoA measurements.
2) **Acoustic Classification:** Each ESP32-S3 node must perform local TinyML-based inference to distinguish target signals (e.g., human-induced knocks) from ambient environmental noise.
3) **Temporal Synchronization:** The system must synchronize all four nodes to a global time-base using the Pulse-Per-Second (PPS) signal from the Bordoza GP-02 GNSS modules.
4) **Centralized Multilateration:** The NVIDIA Jetson Nano gateway shall compute a 2D location fix by correlating a minimum of three synchronized timestamps received via LoRa.
5) **Real-time Visualization:** The estimated coordinates must be transmitted to a web-based dashboard for immediate geographical visualization.

B. Non-Functional Requirements

1) **Latency / Real-time:** The end-to-end latency, from the occurrence of the acoustic event to the coordinate display on the dashboard, must not exceed 2 seconds to support rapid search-and-rescue operations.
2) **Precision:** To maintain sub-meter localization accuracy, each node must provide a synchronization accuracy within $10\mu s$ relative to the Bordoza GP-02 master clock.
3) **Reliability:** The communication link must remain stable in obstructed environments; the E22-900T22D LoRa

modules must maintain a packet delivery ratio (PDR) of at least 95% within the targeted deployment radius.

4) **Scalability:** The gateway architecture must be capable of handling asynchronous data bursts from up to 8 nodes without significant processing jitter.
5) **Energy Efficiency:** Each sensor node must be optimized for low power consumption, enabling at least 24 hours of continuous operation on a standard 1000mAh LiPo battery.
6) **Security:** Data packets transmitted over LoRa must include a unique node identifier and a CRC check to prevent the processing of corrupted or spoofed timestamp data. CRC check feature is already included in LoRa communication.
7) **Acoustic Classification:** Each ESP32-S3 node must perform local TinyML-based inference to distinguish target signals (e.g., human-induced knocks) from ambient environmental noise. While the system can perform localization without ML, this module is essential for reducing false positives and ensuring reliability in high-noise disaster environments.

C. Timing and Execution Model

The system operates on an event-driven execution model. While the GNSS synchronization task runs periodically to maintain clock alignment, the localization pipeline is dormant until an acoustic signal exceeds a threshold. Upon detection, the nodes transmit metadata asynchronously. The gateway implements a "wait-for-quorum" logic, where TDoA calculations begin once a specific number of timestamps are received within a defined temporal window.

D. Assumptions and Design Limitations

- **Synchronization:** It is assumed that all four nodes have sufficient GNSS lock to receive the PPS signal from their respective GP-02 modules.
- **Medium:** The propagation environment is assumed to be homogeneous, allowing for a standard speed-of-sound constant in the TDoA equations.

E. Resource Model

- **Memory:** ESP32-S3 nodes utilize DMA buffers to handle I2S audio streams without CPU overhead, preserving RAM for Edge AI.
- **Bandwidth:** The LoRa star topology is optimized for low duty cycles; each node transmits only small packets, preventing channel saturation even during frequent acoustic triggers.

VI. Methodology

This section presents the design and implementation of the proposed SAR-Acoustic Localization system. The objective of the system is to accurately localize acoustic sources in disaster environments using a distributed, low-cost embedded architecture.

A. System Architecture and Implementation

The proposed system consists of multiple spatially distributed sensor nodes and a central processing unit. Each sensor node is equipped with a microphone, an embedded microcontroller (ESP32), and a GNSS module to enable precise time synchronization.

Each node continuously acquires acoustic signals using its onboard microphone at a fixed sampling rate to ensure temporal consistency across the network. To achieve accurate localization, all nodes are synchronized using GNSS Pulse-Per-Second (PPS) signals, allowing precise timestamping of detected events.

To reduce communication overhead, the system employs LoRa-based transmission, where only essential event information—specifically timestamp differences and node identifiers—is transmitted instead of raw audio data. This approach significantly reduces bandwidth usage and power consumption.

At the node level, a lightweight event detection mechanism is implemented. Incoming audio segments are first evaluated using an energy-based thresholding method. If the signal energy exceeds a predefined threshold, a classification module is invoked to determine whether the signal corresponds to a potential survivor-related acoustic event. Upon detection, the event is timestamped relative to the latest PPS signal and transmitted to the central unit.

The central processing unit, implemented on an NVIDIA Jetson Nano platform, aggregates data from all sensor nodes, aligns timestamps, and performs source localization using an optimization-based approach.

B. Localization via Correlation-Based Hyperbolic Positioning

The localization process is based on Time Difference of Arrival (TDoA) estimation using cross-correlation between signals acquired at spatially distributed sensor nodes.

For each detected acoustic event, the corresponding signal segments from different nodes are first aligned in time. Pairwise cross-correlation is then computed to estimate the relative time delays between nodes. The time delay Δt_{ij} between node i and node j is obtained by identifying the lag that maximizes the cross-correlation function.

Given the estimated time differences, the source location is determined using hyperbolic positioning. For a pair of nodes located at positions $\mathbf{p}_i$ and $\mathbf{p}_j$, the set of possible source locations satisfies:

$$\|\mathbf{x} - \mathbf{p}_i\| - \|\mathbf{x} - \mathbf{p}_j\| = c \cdot \Delta t_{ij} \tag{1}$$

where $\mathbf{x}$ is the unknown source position and c denotes the speed of sound.

Each pairwise TDoA measurement defines a hyperbola on which the source must lie. By combining multiple such hyperbolic constraints from different node pairs, the source position is estimated as the intersection point of these curves.

To obtain a robust solution in the presence of noise, the localization problem is formulated as a nonlinear least squares optimization:

$$\min_{\mathbf{x},\mathbf{y}} \sum_{(i,j)} \left(\|\mathbf{x} - \mathbf{p}_i\| - \|\mathbf{x} - \mathbf{p}_j\| - c \cdot \Delta t_{ij}\right)^2 \tag{2}$$

Prior to optimization, unreliable time delay estimates are filtered based on correlation peak quality. The resulting solution provides the estimated 2D coordinates of the acoustic source.

C. Numerical Example under Statistical Noise

To illustrate the robustness of the Nonlinear Least Squares (NLLS) formulation, consider a 2D spatial setup with three sensor nodes located at $\mathbf{p}_1 = (0, 0)$, $\mathbf{p}_2 = (4, 0)$, and $\mathbf{p}_3 = (2, 3)$ meters. The true acoustic source is positioned at $\mathbf{x}_{\text{true}} = (1.5, 1.8)$ meters. Assuming a constant speed of sound $c = 343$ m/s, the theoretical distances $d_i = \|\mathbf{x}_{\text{true}} - \mathbf{p}_i\|$ are:

$$d_1 = 2.343 \text{ m}, \quad d_2 = 3.081 \text{ m}, \quad d_3 = 1.300 \text{ m} \tag{3}$$

In a practical environment, cross-correlation estimates are corrupted by reverberation and ambient noise. To model this, zero-mean Gaussian noise $\mathcal{N}(0, \sigma^2)$ with a standard deviation of $\sigma = 0.5$ ms is added to the theoretical TDoA values. The resulting measured delays ($\Delta t_{ij}^{\text{meas}}$) are summarized in Table II.

TABLE II
TRUE VS. MEASURED TDOA WITH GAUSSIAN NOISE ($\sigma = 0.5$ MS)

Node Pair	True TDoA (ms)	Measured TDoA (ms)	Error (ms)
1 vs. 2 (Δt_{12})	-2.150	-1.902	+0.248
3 vs. 2 (Δt_{32})	-5.191	-5.260	-0.069
1 vs. 3 (Δt_{13})	+3.041	+3.365	+0.324

Due to the measurement error, the hyperbolic curves defined by these TDoAs no longer intersect at a single distinct point. Applying the NLLS optimization over the noisy measurements yields an estimated source position of $\hat{\mathbf{x}} = (1.55, 1.83)$ meters. Despite the presence of measurement noise, the formulation successfully converges to a highly accurate coordinate, minimizing the overall statistical error across all sensor pairs.

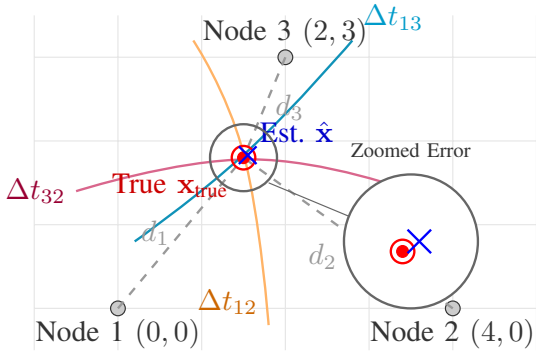

Fig. 3. Visual representation of the numerical example. Measurement noise causes a slight deviation between the true source $\mathbf{x}_{\text{true}}$ (red circle) and the NLLS estimated source $\hat{\mathbf{x}}$ (blue cross). The three hyperbolic curves (orange Δt_{12}, purple Δt_{32}, cyan Δt_{13}) no longer intersect at a single point. The zoomed inset illustrates the spatial error resulting from the $\sigma = 0.5$ ms TDoA noise.

D. GNSS Deployment Strategy and Signal Reliability

A critical challenge in post-disaster scenarios is GNSS signal attenuation due to structural debris. To maintain the 10 μs synchronization precision required for TDoA, the system utilizes a *periphery-anchored deployment model*. In this strategy, sensor nodes are positioned at stable, surface-level access points or along the perimeter of the collapse zone where a clear line-of-sight to the sky is preserved. While the nodes remain at the surface to ensure continuous GNSS lock and PPS reception, their high-sensitivity I2S microphones are oriented toward the debris. This configuration assumes that while the node requires satellite visibility for timing, the acoustic signals from trapped survivors can still propagate through the rubble to the surface-mounted sensors. This deployment model ensures that the system operates with a global time-base without requiring deep-rubble signal penetration for the synchronization hardware.

E. Edge Node / Jetson Operations

The operation of each edge node is summarized in Algorithm 1. The algorithm describes the process of continuous audio monitoring, event detection, classification, timestamping, and communication. By performing early-stage filtering and classification at the edge, the system minimizes unnecessary data transmission while maintaining responsiveness to relevant acoustic events.

Algorithm 1 Edge Node Event Detection, Timing, and LoRa Uplink

Require: Audio stream, GNSS PPS, node ID n, LoRa link
Ensure: Trusted event packet uplink for target acoustic events
Initialize microphone DMA, STA/LTA detector, FFT prefilter, ML model
Initialize GNSS-disciplined clock and LoRa transceiver
while node is active **do**
  Acquire next audio chunk $\mathbf{c}_k$
  $d \leftarrow$ DetectTrigger($\mathbf{c}_k$)
  if $d = \varnothing$ **then**
    continue
  end if
  $f \leftarrow$ FFTAnalyze(d.trigger_chunk)
  $m \leftarrow$ Classify(d.trigger_chunk, f)
  if RESCUERULE(m, f, d.peak) is true **then**
    Force m to target class
  end if
  if m.action $\neq$ target **then**
    Store local reject/log entry and continue
  end if
  $t_\mu \leftarrow$ DisciplinedClockMicros()
  Build packet $(n, t_\mu, \text{mag}, f_0, \text{class}, \text{conf}, \text{snr}, \text{battery}, \ldots)$
  Append CRC and optional authentication tag
  Transmit via LoRa with bounded retry policy
  if transmission fails **then**
    Queue compact event summary for delayed resend
  end if
end while

Algorithm 2 Production Web Backend and Operations Dashboard Flow

Require: Localization events, authenticated users, persistent storage
Ensure: Real-time situational awareness and incident traceability
while backend service is active **do**
  Receive localization event from central service/message bus
  Validate schema, timestamp freshness, and source identity
  Persist event, packet summary, and quality metrics in database
  Update active incident state and node-health aggregates
  Push incremental updates to clients via WebSocket/SSE
  if event status is CONFIRMED and severity $\geq$ threshold **then**
    Create alert record and notify responders
  end if
end while

Dashboard client loop:
while operator session is active **do**
  Subscribe to live stream and fetch latest incident snapshot
  Render map marker, timeline, status badges, and node telemetry
  Allow operator acknowledge/escalate/close actions with audit log
end while

Algorithm 3 Central Service Event Assembly and TDoA Localization

Require: Uplink packets $\mathcal{P}$, node positions $\mathbf{p}_i$, sound speed c
Ensure: Estimated source position, quality metrics, and decision status

```
Validate packet integrity/authentication and drop invalid packets
Group packets by event-time proximity until timeout
if |P| < N_min then
    return status WEAK_DATA
end if
if PhysicalConsistencyCheck(P) fails then
    return status REJECTED
end if
Select reference node r (earliest timestamp, quality tie-break)
for each node i in P do
    Δt_i ← t_i − t_r
    Δd_i ← c Δt_i
    w_i ← sqrt((snr_i · conf_i)/100)
end for
Solve multilateration with robust pipeline:
    Chan–Ho → Taylor/robust LSQ/LM (NM fallback)
Compute GDOP, residual, confidence ellipse, and range checks
Apply anti-ghost and decision-score filters
Assign final status in {CONFIRMED, WEAK_DATA, OUT_OF_BOUNDS, REJECTED}
Publish result to operations backend and alerting queue
return (x̂, status, diagnostics)
```

Algorithms 2 and 3 describe the central processing and visualization stages of the system. The central unit computes TDoA values and estimates the source position using hyperbolic localization, while the web server provides real-time monitoring and visualization of detected events. This separation of edge, processing, and visualization layers ensures scalability and efficient system operation.

F. Meaning of Chan–Ho, Taylor, LSQ, and LM in TDoA Localization

For a source at unknown position $\mathbf{x}$ and a reference node r, each time-difference-of-arrival (TDoA) measurement is modeled as

$$z_i = c\,(t_i - t_r) = \|\mathbf{x} - \mathbf{p}_i\|_2 - \|\mathbf{x} - \mathbf{p}_r\|_2 + n_i, \qquad (4)$$

where $\mathbf{p}_i$ is node position, c is sound speed, and n_i is measurement noise. The central service pipeline in Algorithm 3 combines the following estimators.

a) Chan–Ho (closed-form initializer).

The Chan–Ho method is a two-stage weighted least-squares (WLS) estimator for hyperbolic localization. It transforms the nonlinear TDoA equations into an approximately linear system, solves for an initial source estimate, and then enforces a geometric constraint in a second step. Its main role is to provide a fast, low-cost initial guess close to the global solution under moderate Gaussian noise.

b) Taylor-series refinement (iterative linearization).

Starting from $\mathbf{x}_k$ (typically from Chan–Ho), the measurement function is linearized using first-order Taylor expansion:

$$\mathbf{h}(\mathbf{x}) \approx \mathbf{h}(\mathbf{x}_k) + \mathbf{J}_k\,\Delta\mathbf{x}, \qquad (5)$$

with residual $\mathbf{r}_k = \mathbf{z} - \mathbf{h}(\mathbf{x}_k)$. The update is obtained by WLS:

$$\Delta\mathbf{x} = (\mathbf{J}_k^T\mathbf{W}\mathbf{J}_k)^{-1}\mathbf{J}_k^T\mathbf{W}\mathbf{r}_k, \qquad (6)$$

then $\mathbf{x}_{k+1} = \mathbf{x}_k + \Delta\mathbf{x}$. This stage improves accuracy when the initialization is already near the true position.

c) LSQ / robust LSQ (objective-level fitting).

Least-squares (LSQ) solves

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \sum_i w_i \left(z_i - h_i(\mathbf{x})\right)^2. \qquad (7)$$

In robust LSQ, the quadratic penalty is replaced by an M-estimator (e.g., Huber) to reduce the influence of outliers and non-line-of-sight (NLoS) errors:

$$\hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \sum_i \rho\left(\frac{z_i - h_i(\mathbf{x})}{\sigma_i}\right). \qquad (8)$$

d) Levenberg–Marquardt (LM).

LM is a damped nonlinear least-squares method that interpolates between Gauss–Newton and gradient descent:

$$(\mathbf{J}^T\mathbf{W}\mathbf{J} + \lambda\mathbf{I})\,\Delta\mathbf{x} = \mathbf{J}^T\mathbf{W}\mathbf{r}. \qquad (9)$$

Large λ improves stability when the model is poorly conditioned; small λ gives faster Gauss–Newton-like convergence near the optimum.

e) NM fallback (Nelder–Mead).

If Jacobian-based iterations fail (e.g., severe nonlinearity, poor conditioning, or unstable derivatives), a derivative-free Nelder–Mead simplex search can be used as a fallback to obtain a feasible local minimum.

f) Why this stack is used in deployment.

The combined strategy "Chan–Ho → Taylor/robust LSQ/LM (NM fallback)" balances speed, numerical stability, and outlier tolerance: Chan–Ho provides a reliable initial point, iterative LSQ/LM improves precision, robust loss limits bad packets, and NM protects against hard failure modes.

g) Quality metrics after solving.

After obtaining $\hat{\mathbf{x}}$, the service computes residual norms, physical range checks, and geometric dilution of precision (GDOP), commonly approximated as

$$\mathrm{GDOP} \approx \sqrt{\mathrm{trace}((\mathbf{J}^T\mathbf{W}\mathbf{J})^{-1})}, \qquad (10)$$

which supports final status assignment (e.g., CONFIRMED, WEAK_DATA, OUT_OF_BOUNDS, REJECTED).

VII. Experimental Setup

The experimental evaluation of the SAR-Acoustic Localization system is designed to validate its localization accuracy, end-to-end latency, edge classification performance, and overall reliability in simulated disaster environments.

A. Platform and Tools

The hardware deployment consists of four distributed sensor nodes and one central gateway. Each sensor node is built around an **ESP32-S3** microcontroller, chosen for its dual-core architecture and vector instructions suitable for Edge AI inference. Acoustic data is captured using an **INMP441** omnidirectional I2S digital microphone. For microsecond-level synchronization, each node is interfaced with a **Bordoza GP-02** GNSS module, utilizing its Pulse-Per-Second (PPS) output triggered via hardware interrupts. Communication is handled by **Ebyte E22-900T22D** LoRa modules operating at 868 MHz.

The central gateway operates on an **NVIDIA Jetson Nano 4GB**, equipped with a matching LoRa receiver to serve as the system's primary edge hub. In this capacity, the Jetson Nano is responsible for radio ingest, event fusion, localization inference, quality control, and the operator interface. Specifically, the gateway, receives LoRa packets from distributed ESP32 sensor nodes, where each packet contains event-driven timestamps and extracted audio features; groups packets by event windows to perform **TDoA localization** using a robust optimization pipeline (utilizing the L-BFGS-B algorithm via **SciPy**) to estimate source positions; computes quality metrics, such as residual errors and Geometric Dilution of Precision (GDOP), to identify and reject unreliable solutions; stores and serves results through a local web API for rescue teams. Software-wise, the ESP32-S3 firmware integrates **TensorFlow Lite for Microcontrollers** for local acoustic classification, while the Jetson Nano utilizes a **Node.js** web server to provide a real-time, **React**-based dashboard for field operators.

The physical experiments are conducted in an outdoor 50 m $\times$ 50 m test field to ensure clear sky visibility for the GNSS modules.

B. Datasets and Workloads

To evaluate the Edge AI classification module, the open-source ESC-50 dataset was utilized. The dataset consists of 2,000 labeled environmental audio recordings balanced across 50 distinct semantic classes. To simulate a search-and-rescue workload, the classes were partitioned into:

- **Target Signals (Survivors):** Audio classes representative of human presence, activity, or distress, such as whistle, knocking.
- **Noise Signals:** Ambient environmental and urban noise classes, including wind, rain, heavy engine sounds, and sirens, used to evaluate the background noise rejection capabilities of the classifier.

For the system-level localization evaluation, physical tests involve the playback of these selected ESC-50 audio samples via an acoustic emitter at predefined ground-truth coordinates (x, y) within the sensor grid. The workloads are divided into two scenarios: *Line-of-Sight (LoS)* in an open field to establish baseline accuracy, and *Non-Line-of-Sight (NLoS)* where concrete blocks and wooden pallets are placed between the audio source and sensors to introduce realistic signal attenuation and multipath reverberation.

C. Evaluation Metrics

The proposed system is compared against a baseline approach consisting of a standard Wi-Fi-based network utilizing generic network time protocols (NTP) without precise PPS synchronization, and standard cross-correlation without Phase Transform (CC) [1]. The evaluation utilizes the following metrics:

- **Localization Accuracy:** Measured using the Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) between the estimated source coordinates and the physical ground-truth position (in meters).
- **End-to-End Latency:** The time elapsed (in seconds) from the acoustic event generation to the coordinate update on the web dashboard.
- **Synchronization Drift:** The maximum measured temporal deviation (μs) between the PPS hardware interrupts across the four deployed nodes.
- **Edge AI Performance:** Evaluated using Accuracy, Precision, Recall, and F1-Score for the TinyML acoustic event classifier running on the ESP32-S3.
- **Network Reliability:** Evaluated via Packet Delivery Ratio (PDR), representing the percentage of successfully received LoRa metadata packets at the gateway within the 2-second temporal window.
- **Energy Consumption:** The average power draw (in milliwatts) of the ESP32-S3 nodes during deep-sleep monitoring, active inference, and LoRa transmission phases.

VIII. System-Level Considerations

Deploying a Search and Rescue (SAR) acoustic localization system in post-disaster environments introduces strict operational requirements. This section outlines the critical system-level considerations, detailing both currently implemented solutions and limitations reserved for future work.

A. Real-Time Constraints and Latency

In SAR operations, minimizing the time between an acoustic event (e.g., a survivor knocking) and the coordinate estimation is critical.

- **Implemented:** By migrating the acoustic classification to the extreme edge (ESP32-S3 via TensorFlow Lite for Microcontrollers), the system avoids transmitting heavy raw audio files. The inference takes only a few milliseconds, and the payload transmitted over LoRa is reduced to lightweight metadata (timestamp, classification label, and node ID).
- **Limitations & Future Work:** LoRa's low bandwidth and regulatory duty-cycle limitations mean that in a scenario with continuous, overlapping noise events, packet queuing could introduce latency. Future work will explore dynamic spreading factor (SF) allocation to optimize airtime.

B. Reliability and Fault Tolerance

Disaster zones are characterized by collapsed infrastructure, making traditional communication networks (Wi-Fi/Cellular) highly unreliable.

- **Implemented:** The use of 868 MHz LoRa modules ensures high-penetration, long-range communication through concrete rubble (Non-Line-of-Sight conditions). Furthermore, the GNSS PPS synchronization ensures that even if a node temporarily loses connection to the gateway, its internal timer remains microsecond-accurate for future Time Difference of Arrival (TDOA) calculations.
- **Limitations & Future Work:** The current architecture relies on a star topology centered around a single Jetson Nano gateway, representing a single point of failure.

Implementing a mesh-routing protocol among the ESP32-S3 nodes to relay packets if the gateway is obstructed is a planned future improvement.

C. Energy Efficiency

Sensor nodes must operate for extended periods (e.g., 48 to 72 hours) on battery power during the critical SAR window.

- **Implemented:** The system heavily utilizes the ESP32-S3's power management features. The I2S microphone is buffered, and the main dual-core processor is kept in a low-power state until a basic amplitude threshold interrupt triggers the Edge AI classification sequence.
- **Limitations & Future Work:** Continuous GNSS module operation for PPS polling draws significant current. Future iterations will duty-cycle the GNSS module alongside an ultra-precise Real-Time Clock (RTC) or integrate compact solar-harvesting circuits to extend operational lifespans.

D. Security Considerations

While SAR operations are emergency-focused, unauthenticated signals could lead to false coordinate generation, misdirecting rescue teams.

- **Implemented:** Currently, security is handled at a basic physical level by utilizing a proprietary LoRa channel and specific packet formatting to filter out generic radio noise.
- **Limitations & Future Work:** Robust cryptographic security is not fully implemented in the current prototype to minimize overhead. Future deployments will mandate AES-128 encryption for all LoRa payloads and cryptographically sign timestamps to prevent replay or spoofing attacks.

E. Scalability

As the disaster area expands, the system must accommodate a denser grid of sensor nodes to maintain localization accuracy.

- **Implemented:** The TDOA mathematical model on the Jetson Nano gateway is designed to accept an arbitrary N-number of nodes. As long as at least three nodes report the same acoustic event (correlated via PPS timestamps), the non-linear least squares algorithm can resolve the 2D coordinates.
- **Limitations & Future Work:** Scaling up to dozens of nodes on a single LoRa channel will result in packet collisions. Transitioning from raw LoRa to a full LoRaWAN MAC implementation with Time-Division Multiple Access (TDMA) slotting is required to support a massive-scale sensor deployment.

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References

[1] Z. Dong and M. Yu, "Research on tdoa based microphone array acoustic localization," in *IEEE International Conference on Electronic Measurement Instruments*, 2015, pp. 1077–1081.

[2] M. Cobos, F. Antonacci, A. Alexandridis, A. Mouchtaris, and B. S. Lee, "A survey of sound source localization methods in wireless acoustic sensor networks," *Wireless Communications and Mobile Computing*, vol. 2017, p. 3956282, 2017.

[3] A. Patwari *et al.*, "Microphone array and raspberry pi interfacing for real-time doa estimation," *International Journal of Emerging Trends in Engineering Research*, vol. 8, no. 5, p. 2020.

[4] J. Yin and M. Verhelst, "Cnn-based robust sound source localization with srp-phat for the extreme edge," *ACM Transactions on Embedded Computing Systems*, vol. 22, no. 3, 2023.

[5] J. Wang, Z. Cheng, L. Jing, and T. Yoshida, "Design of a 3d localization method for searching survivors after an earthquake based on wsn," pp. 221–226, 2011.

[6] K. J. Piczak, "ESC: Dataset for Environmental Sound Classification," in *Proceedings of the 23rd ACM international conference on Multimedia*, 2015, pp. 1015–1018.